

Heart route rescue

Teacher guide and worked answers

Trace blood through the heart, lungs and body and explain the heart's pumping job.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | B33:C33

Describe the human circulatory system and the job of the heart.

Directly develops this stated outcome with secondary-age tasks at an upper-KS2 conceptual level. Support and stretch are flexible; this exemplar is not a claim of qualification approval.

Prior learning

Distinguish digestive and circulatory jobs.

Know that tissues need oxygen.

Success evidence

Trace body → right heart → lungs → left heart → body.

Explain that arteries carry blood away and veins return it.

Correct a reversed arrow or oxygen-colour misconception.

Equipment

Shared route sheet, pencils, optional tokens moved on the desk.

Preparation

Print selected routes.

Explain that right/left refer to the person's heart; the route map is not an anatomical front view.

Practical care

Use desk tokens; no pulse measurements, exercise, bodily samples or physical restraint.

Do not infer a pupil's health from participation or prior knowledge.

Ready to teach alternative

Desk route rescue uses supplied labels and arrows; no practical physiological measurements.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The courier has lost the route
2	Retrieval	2-5	Which system does the delivery?
3	I do	5-13	I trace one complete journey

Slide	Stage	Minutes	Focus
4	I do	Within stage	I name the vessel by direction
5	I do	Within stage	I explain the pump
6	We do	13-21	Route rescue: four delivery links
7	We do	Within stage	Find the two broken labels
8	We do	Within stage	An oxygen-poor artery?
9	Hinge	21-24	Blood leaves the lungs. Where next?
10	You do	24-34	Build a reliable route map
11	You do	Within stage	Read your route aloud or silently
12	You do	Within stage	Check the key
13	Review	34-38	Test another map
14	Review	Within stage	Choose a model improvement
15	Exit	38-40	A route in three answers

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The courier has lost the route

Pupils trace an imagined token at a desk. The model represents blood flow, not a blood cell choosing where to go.

2 Retrieval Which system does the delivery?

R1/R2 blood. R3 oxygen, nutrients or water. Establish that lungs supply oxygen to blood, not the heart.

3 I do I trace one complete journey

Think aloud: I follow arrows, not colour alone. Right heart sends blood to lungs. Blood gains oxygen there; returns to left heart; is pumped to body. From tissues it returns to right heart. This is a route map, not anatomical position. Oxygen-poor still contains some oxygen.

4 I do I name the vessel by direction

Shared I do time. Think aloud: Heart-to-lungs is away, so it is an artery even though oxygen-poor. Lungs-to-heart is towards, so a vein even though oxygen-rich. The visual shows one outward route and names the return.

5 I do I explain the pump

Shared I do time. Trace forward flow through open valve. On the second panel show attempted reverse direction meeting closed leaflets. This is a schematic teaching model, not a surgical drawing.

6 We do Route rescue: four delivery links

A right heart → lungs; B lungs → left heart; C left heart → body; D body → right heart. Desk cards or spoken sequence. Ask pupils to identify where oxygen enters and where it is used.

7 We do Find the two broken labels

Shared We do time. Fault1 requires lungs → left heart → body. Fault2 blood is red; veins lead towards heart and can carry oxygen-rich blood, e.g. pulmonary veins. Use two separate reasons.

8 We do An oxygen-poor artery?

Shared We do time. It is the pulmonary artery, carrying oxygen-poor blood away. Blue is only a key. Do not require pulmonary for supported route.

Reveal: Artery: it goes away from the heart. Blue means oxygen-poor in this model.

9 Hinge Blood leaves the lungs. Where next?

A is incorrect: blood passes through the left heart before the body. B is correct. C is incorrect: the right heart sends blood to the lungs. Re-trace the lung route if responses are split.

A. Straight to the body, bypassing the heart. - Blood returns to the left heart before being pumped to the body.

B. To the left side of the heart. - The left side receives oxygen-rich blood from the lungs.

C. To the right side of the heart. - The right side receives blood returning from the body.

10 You do Build a reliable route map

10 minutes across You do. All routes assess circulation, direction and pumping. Differentiate recording, not the core idea.

11 You do Read your route aloud or silently

Shared You do time. Adult may point to a gap without providing the destination. Tokens need not physically move around the room.

12 You do Check the key

Shared You do time. Labels remain useful in monochrome printing. Avoid colour alone as the indicator.

13 Review Test another map

4 minutes across Review. Teacher offers anonymous or private review. Accept a pupil choosing teacher review over peer exchange.

14 Review Choose a model improvement

Lundy: collect choices, pick one detail to draw next on the board and explain which pupil reasoning influenced it. Capillaries clarify exchange; chambers clarify receiving and pumping.

15 Exit A route in three answers

E1: left side of the heart. E2: carries blood away from the heart. E3: contracting heart muscle. E4 on the pupil page explains the blue convention. Collect answers before a public explanation.

Answers and assessment

R1-R3

R1/R2 blood. R3 oxygen, nutrients or water are acceptable. Oxygen enters blood in the lungs.

W1-W3

W1 A → B → C → D → A; oxygen enters in lungs. W2 lungs → left heart → body; blood is red, and veins are defined by flow towards the heart. W3 pulmonary artery, because flow is away from the heart; oxygen-poor blue is a diagram convention.

Hinge A-C

A incorrect: left heart must come before body. B correct. C incorrect: right heart receives blood from body and sends it to lungs.

S1-S6

S1 right heart → lungs → left heart. S2 lungs. S3 artery. S4 muscles/blood/backward. S5 blue means oxygen-poor; human blood is red. S6 complete directed loop: right heart → lungs → left heart → body → right heart. Accept any spatial arrangement with correct arrows.

N1-N4

N1 correct loop; oxygen gain at lungs. N2 artery/vein/artery/vein; away versus towards heart. N3 muscle contractions pump blood; valves resist backward flow. N4 blue labels oxygen-poor blood on this diagram; it is not actual blue blood.

T1-T4

T1 right heart → lungs → left heart → body → right heart. T2 pulmonary artery goes away from right heart towards lungs and carries oxygen-poor blood. T3 muscle contraction supplies force; valves help make flow one-way. T4 limits include missing four chambers, capillary exchange detail, vessel branching, actual shapes and relative sizes. A justified useful addition such as capillaries showing oxygen exchange is sufficient.

E1-E4 / reflection

E1 left side of heart. E2 carries blood away from heart. E3 contracting heart muscle. E4 oxygen-poor; blood is red. Reflection needs no particular personal response.

Assessment and re-teach

Secure: complete route with both heart sides, direction-based vessel definition and pumping-muscle explanation. Re-teach skipped-heart errors with one token traced twice through labelled heart sides. Re-teach blue-blood or all-arteries-oxygen-rich errors by comparing the pulmonary artery with the colour key.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

No pupil pulse readings or exercise comparison is part of this lesson. Right and left are the person's anatomical sides, not necessarily the left and right of a page.

For stretch, name atria as receiving chambers and ventricles as pumping chambers if already familiar. Do not let optional chamber names replace accurate core route reasoning.

Blood always remains red in this modelled human system. Oxygen-poor is a relative label, not an assertion of zero oxygen.

Interactive We do uses predict, test and explain; paper cards W1-W3 provide the same decisions. A desk token can trace the model without an animated screen. Movement is optional and pupils control the pace.

Access and responsive teaching

Supported

Use numbered route words and direction choices.

Standard

Trace both loops and justify artery/vein names.

Stretch

Repair a faulty full route and evaluate model limitations.

Regulation

Offer quiet pointing, private writing, drawing or adult scribing. Pass-and-return for public questions. Use fictional cases; no personal medical disclosure or comparison of bodies. Assign response routes from current evidence, not fixed pupil labels.

Misconceptions to check

Blue blood flows in veins.

Human blood is red; blue in this model means oxygen-poor.

Check: Ask what the blue line stands for.

Every artery is oxygen-rich.

Arteries are defined by carrying blood away from the heart; the pulmonary artery carries oxygen-poor blood.

Check: Ask the direction of blood in the heart-to-lungs route.

Blood travels heart → body → lungs → heart.

In a complete circuit it passes through the heart between the lungs and body.

Check: Trace body → right heart → lungs → left heart → body.

Word	Meaning
heart	A muscular organ that pumps blood.
artery	A blood vessel carrying blood away from the heart.
vein	A blood vessel carrying blood towards the heart.
capillary	A very small vessel where substances pass between blood and tissues.
oxygen-rich	Containing more oxygen than the returning oxygen-poor blood.
valve	A structure that helps prevent backward flow.

Scientific references

[NHLBI: How blood flows through the heart](#)

Checked September 2026. Right-heart/lung/left-heart/body sequence and oxygenation. No adult pulse thresholds are used for pupils.

[NHLBI: How the heart beats](#)

Checked September 2026. Heart muscle contraction and valves; pupil health is not assessed from pulse values.